

Load Interconnection System Impact Study Report

For Queue Project Q12-0171

Parsons Tap 138 kV

January 17, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0171. The Interconnection Customer proposes a 557.6 MW large load (data center) facility interconnecting at the Parsons Tap 138 kV substation in the Virginia area. The request was received on November 03, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 129 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$165,583,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mccartney Substation to Parsons Tap 138 kV	Yarnen to Mccartney 345 kV	159.8	480	58.1
Mccartney 138 kV/1 kV Transformer Bank	Mccartney to Yarnen 345 kV	156.7	508	58.1
Deen Point to Locks 138 kV	Mccartney 138 kV/1 kV Transformer	147.5	169	26.3
Villa Del Rey 3010 Substation to St. Joseph 138 kV	Mccartney 138/1 kV Autotransformer	145.9	217	45.5
St. Joseph to Emporia 138 kV L	Mccartney Substation 138/1kV Transformer Bank	138.6	223	45.5
Kewaunee to Tyler 138 kV	Mccartney 138/1 kV Autotransformer	134.1	227	26.3
Goldfield to Kewaunee Substation 138 kV	Mccartney 138/1kV XFMR	131.6	212	26.3
Goldfield to Deen Pt 138 kV	Mccartney 138 kV/1 kV Transformer Bank	130.9	214	26.3
Mccartney 345 kV/1 kV Auto	Mccartney to Yarnen 345 kV line	127.6	624	58.1
Parsons Tap to Flat Top 138 kV	Mccartney Substation 138/1 kV XFMR	127.3	232	45.5
Flat Top to Emporia 138 kV	Mccartney 138/1 kV Transformer	127.0	232	45.5
American to Ridgeley 345 kV	E Devon to Thiele Kaolin Reedy Crk 345 kV	120.8	1,263	7.1
Alligator 500/345kV XFMR	Enka to Thiele Kaolin Reedy Creek 345 kV	120.3	544	5.2
Person to State School 345 kV	Person to Mecklenburg Power Station 345 kV	114.7	1,605	5.0
Rosemary Power Sta to Seven Mile 138 kV	Mccartney 138 kV/1 kV Transformer	112.7	245	21.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
International Paper Valliant Ok to Urban 345 kV	Front Royal to Urban 345 kV	111.4	1,276	7.1
Caon City to Seven Mile 138 kV	Mccartney 138/1 kV XFMR	110.3	242	21.2
Klamath Falls to Bauxite 138 kV	Mecklenburg Power Sta to Kerr Dam 345 kV Line	108.7	253	5.6
Klamath Falls to Kenbridge 138 kV	Mecklenburg Power Sta to Kerr Dam 345 kV	108.3	248	5.6
Spindle to Ridgeley 345 kV	E Evon to Thiele Kaolin Reedy Creek 345 kV	107.1	1,495	6.8
Keokuk to Kenbridge 138 kV	Mecklenburg Power Station to Ker Dam 345 kV	106.7	247	5.6
Floraday to Sonoma 3 138 kV	Mccartney Substation 138/1kV XFMR	106.4	203	26.3
Sonoma 3 to Locks 138 kV	Mccartney 138 kV/1 kV Transformer	106.0	203	26.3
East Longmeadow to Parsons Tap 138 kV	Mccartney Substation 138-1 kV Transformer	102.3	298	54.5
Floraday to Villa Del Rey 3010 138 kV	Mccartney 138/1 kV TX	101.1	268	40.8

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Parsons Tap Substation to Mccartney 138 kV	Rebuild line with higher-capacity conductor	\$3,926,000	92% (\$3,621,000)
Mccartney 138 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$6,599,000	92% (\$6,088,000)
Deen Point to Locks 138 kV	Rebuild line with higher-capacity conductor	\$1,092,000	100% (\$1,092,000)
Villa Del Rey 3010 Substation to St. Joseph 138 kV	Rebuild line with higher-capacity conductor	\$10,229,000	100% (\$10,229,000)
St. Joseph to Emporia 138 kV	Rebuild line with higher-capacity conductor	\$26,713,000	100% (\$26,713,000)
Kewaunee to Tyler 138 kV	Rebuild line with higher-capacity conductor	\$7,059,000	100% (\$7,059,000)
Kewaunee to Goldfield 138 kV	Rebuild line with higher-capacity conductor	\$3,283,000	100% (\$3,283,000)
Deen Pt to Goldfield 138 kV L	Rebuild line with higher-capacity conductor	\$1,796,000	100% (\$1,796,000)
Mccartney 345 kV/1 kV Auto	Replace transformer with higher-rated unit	\$11,875,000	92% (\$10,955,000)
Parsons Tap to Flat Top 138 kV	Rebuild line with higher-capacity conductor	\$8,598,000	100% (\$8,598,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Flat Top to Emporia 138 kV	Rebuild line with higher-capacity conductor	\$3,260,000	100% (\$3,260,000)
American to Ridgeley 345 kV	Rebuild line with higher-capacity conductor	\$25,142,000	16% (\$3,894,000)
Alligator 500/345kV XFMR	Replace transformer with higher-rated unit	\$35,048,000	11% (\$3,924,000)
Person Substation to State School 345 kV Line	Rebuild line with higher-capacity conductor	\$51,742,000	21% (\$10,868,000)
Rosemary Power Sta to Seven Mile 138 kV	Rebuild line with higher-capacity conductor	\$25,169,000	100% (\$25,169,000)
International Paper Valliant Ok to Urban 345 kV	Rebuild line with higher-capacity conductor	\$53,525,000	13% (\$7,097,000)
Seven Mile to Canon City 138 kV	Rebuild line with higher-capacity conductor	\$6,659,000	100% (\$6,659,000)
Bauxite to Klamath Falls 138 kV	Reconductor line	\$2,250,000	60% (\$1,342,000)
Klamath Falls to Kenbridge 138 kV	Reconductor line	\$2,029,000	60% (\$1,210,000)
Spindle to Ridgeley 345 kV Line	Reconductor line	\$17,820,000	16% (\$2,922,000)
Kebridge Substation to Keokuk 138 kV	Reconductor line	\$2,374,000	60% (\$1,416,000)
Sonoma 3 to Floraday 138 kV	Reconductor line	\$4,555,000	100% (\$4,555,000)
Sonoma 3 to Locks 138 kV	Reconductor line	\$1,848,000	100% (\$1,848,000)
Parsons Tap to E Longmeadow 138 kV	Reconductor line	\$9,101,000	100% (\$9,101,000)
Villa Del Rey 3010 to Flraday 138 kV	Reconductor line	\$3,168,000	91% (\$2,884,000)
	Total allocated to project		\$165,583,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.